

Georges Parfait Djimefo Kapen | AI Educator & Data Scientist | Applied ML | Responsible AI

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Profile

AI professional and educator with a PhD in Computer Engineering (defense Summer 2026) and demonstrated experience teaching technical courses at university level, designing practical learning materials, and deploying production AI/ML systems. Research in **multi-agent reinforcement learning (MARL)**, **graph neural networks (GNNs)**, and **zero-trust AI security**. Industry experience deploying **RAG agents**, **LLM pipelines**, **Streamlit dashboards**, and **ML evaluation frameworks (RAGAS)** at Ericsson Canada. Committed to **responsible AI by design**.

Technical Skills

Education Tools: Jupyter Notebooks, Streamlit (interactive demos), Git/GitHub, Python labs

AI / ML: PyTorch, TensorFlow, Scikit-Learn, MARL (QMIX, QTRAN), GNN, DQN, NLP, Computer Vision, LLMs, RAGAS

LLM / Agents: LangChain, LangGraph, LangSmith, MCP, RAG (FAISS), CRAG, Claude (Haiku, Opus), AWS Bedrock, Kiro

Data: Python, Pandas, NumPy, PySpark, Databricks, Snowflake, Delta Lake, SQL

Cloud: AWS (SageMaker, Bedrock, EKS, Lambda, CloudWatch, S3, ECR, CloudFormation), Kubernetes, Docker

DevOps / IaC: CI/CD (GitHub Actions, CodeBuild, OIDC), Git, Jenkins, GitLab, Terraform

Languages: Python, R, Java, C++, SQL, JavaScript, TypeScript

Methods: Responsible AI, MLOps, AI Governance, Agile/Scrum, TDD, Statistical Modeling

Teaching & Education Experience

Polytechnique Montréal

Lecturer — Python Procedural Programming

Montréal
2025–2026

- Teach the Python procedural programming course; communicate complex technical concepts to non-technical audiences.

Polytechnique Montréal

Lecturer — Python Procedural Programming

Montréal
2022–2023

- Teach the Python procedural programming course.

Université de Montréal

Scientific Assistant — AI MOOC

Montréal
2021–2022

- Design courses and practical work in supervised learning (Python).

Protestant University of Central Africa

Associate Professor — Statistics & Probability

Cameroon
2010–2012

- Teach statistics and probability courses.

ISSEA (Sub-regional Institute of Statistics and Applied Economics)

Part-time Lecturer — Data Analysis

Cameroon
2009–2010

- Teach the Data Analysis course (R).

Applied AI & Industry Experience

Ericsson

ML Intern — Agentic AI & Cloud Systems

Montréal
2026

- Built an **executive-facing Streamlit dashboard** exposing model accuracy, latency, usage, and cost-per-query metrics for non-technical leadership decisions.
- Trained a custom **Dual-Attention SLM** achieving **F1 90.6 %**, 99.97 % accuracy, and **1 ms/log** inference latency on AWS SageMaker; evaluated using **RAGAS** faithfulness and relevancy scoring.
- Designed and deployed a **production RAG agentic AI system**: 6-tool ReAct agent (LangChain, LangGraph, Claude Haiku on Bedrock, FAISS) automating telecom radio unit log anomaly detection.
- Implemented a **7-stage CI/CD pipeline** (GitHub Actions + CodeBuild, OIDC) with **131 automated tests**; deployed on AWS EKS (Kubernetes); fully autonomous retraining loop with AI governance and drift monitoring.
- Engineered an **event-driven pipeline** (S3 → Lambda → SQS → K8s worker → S3 → SNS alerts) for high availability.
- Stack:** Python, AWS (SageMaker, Bedrock, EKS, Lambda, SQS, SNS, S3, ECR, CloudFormation, CloudWatch), LangChain, LangGraph, LangSmith, FAISS, MCP, Claude, Kiro, Docker, Kubernetes, Terraform.

Intact Financial Corporation

AI Developer I

Montréal
2022–2024

- Ingest and process data at scale.
- Monitor pipelines and develop automated testing solutions ensuring system reliability.
- *Stack*: PySpark, SQL, Databricks, Snowflake, AWS, Shell, Git.

Publications

- [1] *Energy-Efficient and Near Real-Time Routing Algorithm for Disaster Monitoring in Wireless Sensor Networks*
G. P. Djimefo Kapen, R. Al Mallah, S. Pierre — Polytechnique Montréal (Published)
- Designed **GAPF**, a novel **meta-learning** framework fusing MARL policies (QMIX & QTRAN) via a **GCN encoder** with only **1 473 parameters**.
 - Achieved $\geq 98\%$ packet delivery, $2\text{--}3.5\times$ less energy, $13\times$ less memory than baselines.
 - Oral presentation at **PolyCongrès 2026**, Polytechnique Montréal.
- [2] *Carbon-Aware Autonomous Data Reduction and Self-Healing Routing for 6G-Integrated Disaster Sensor Networks*
G. P. Djimefo Kapen, R. Al Mallah, S. Pierre — Polytechnique Montréal (Submitted)
- Proposed **CALASH**, the **first protocol** jointly optimizing Lifecycle Carbon Intensity via a hybrid **Lyapunov-DQN** engine.
 - Achieved 60 % longer network lifetime and 33 % lower LCI than baselines.
- [3] *Zero-Trust Shielded Graph-Structured Decision Control for Secure Autonomous Recovery in AI-Native 6G Networks*
G. P. Djimefo Kapen, R. Al Mallah, S. Pierre — Polytechnique Montréal (Submitted)
- Designed **CASTER-ZT**, combining **GNN-based** decision making with a formally bounded **zero-trust** admissibility shield.
 - Achieved 74–98 % rogue detection with **zero false positives**, 3 ms latency within O-RAN 10 ms budget.

Selected Projects

- Polytechnique Montréal** Montréal
Integrator Project — IATA 2021
- “Flight Data eXchange — Develop an initial and contextual flight data assessment tool.” (Dash, Python, React, Electron, Axios).
- Polytechnique Montréal** Montréal
NLP Project 2021
- Extraction of keywords from texts as well as their type (Python).
- Polytechnique Montréal** Montréal
Undergraduate Research Initiation Intern 2020–2021
- Voice Recognition modeling for secure bank payments; Deep Learning (Python) deployed on Android (Java).

Education

- Polytechnique Montréal** Montréal, Canada
PhD in Computer Engineering Summer 2026
- AI-driven routing, carbon-aware networking, and zero-trust security for 6G sensor networks. First paper published.
- Polytechnique Montréal** Montréal, Canada
Engineering Diploma in Software Engineering
- Concentration: Artificial Intelligence — Data Sciences.
- Advanced National School of Engineering (ENSPY)** Yaoundé, Cameroon
Master’s Degree in Applied Statistics
- Equivalency: Master’s in Probability — Statistics.
- University of Yaoundé I** Yaoundé, Cameroon
Master’s Degree in Mathematics
- Equivalency: Specialized Higher Studies Diploma (DESS) in Mathematics.

Licenses & Certifications

- AWS Certified Cloud Practitioner (CLF-C01)
Certification Prep Path (Pluralsight), Montréal
- AWS Certified Developer — Associate (DVA-C02)
Certification Prep Path (Pluralsight), Montréal

Conferences & Seminars

- PolyCongrès 2026, Polytechnique Montréal** Montréal, Canada
Oral Presentation Mar–Apr 2026
- “Energy-Efficient and Near Real-Time Routing for Disaster Monitoring in Wireless Sensor Networks — Graph-Attentive Policy Fusion (GAPF).”

Volunteering

- Volunteer at the IEEE branch of Polytechnique Montréal, Montréal, 2018.
- Volunteer at the ConFoo conference, Montréal, 2018.